

*Anaeramoeba flamelloides: An Anaerobic Amoeba with an Unusual  
Symbiotic Partnership*

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In shallow marine sediments worldwide, *Anaeramoeba flamelloides* inhabits anoxic environments in marine sands and muds. *A. flamelloides* is a free-living anaerobic marine amoeboflagellate belonging to Metamonada, primarily exhibiting an amoeboid form but capable of transitioning to a flagellate form with two or four thickened, isokont flagella. SSU rRNA gene sequencing could not robustly resolve the relationships of these organisms to other known eukaryotic lineages, contributing to their classification within the newly established family Anaeramoebidae. The genus *Anaeramoeba* contains six species, with *A. flamelloides* designated as the type species (Táborský et al. 2017).

*A. flamelloides* is a microscopic amoeboflagellate whose morphology varies with cellular activity. Its active, crawling form is fan-shaped, with a broad, flattened leading edge and delicate trailing projections, while non-motile or floating cells become less polarized or more compact (Táborský et al. 2017). Internally, *A. flamelloides* has a complex cellular architecture, including a prominent central nucleus and a large ventral acentriolar centrosome from which microtubules radiate throughout the ventral side of the cell (Táborský et al. 2017; Jerlström-Hultqvist et al. 2024). The cells also contain hundreds of double-membrane-enclosed hydrogenosomes tightly associated with a mass of symbionts near the nucleus (Jerlström-Hultqvist et al. 2024). Two reconstructed cells contained 374 and 307 hydrogenosomes, accounting for approximately 6% of total cell volume (Jerlström-Hultqvist et al. 2024).

Using deep transcriptomic sequencing and a phylogenomic dataset comprising 95 taxa, 155 proteins, and 53,637 sites, researchers found that Anaeramoebae, the phylum-level lineage containing *A. flamelloides*, forms a distinct branch within Metamonada, sister to Parabasalia (Stairs et al. 2021). Adding Anaeramoebae to the phylogeny helped break up the long branch leading to parabasalids, reducing long-branch attraction that had previously complicated relationships among major metamonad groups (Stairs et al. 2021). This relationship is particularly surprising because *A. flamelloides* is predominantly amoeboid with sparse microtubules radiating from an acentriolar centrosome, whereas most parabasalids are flagellates with elaborate cytoskeletons (Stairs et al. 2021). Like other metamonads, *A. flamelloides* possesses mitochondrion-related organelles (MROs) rather than classical aerobic mitochondria.

However, its hydrogenosomes retain mitochondrial features not previously observed in other metamonad lineages, providing insight into the evolutionary transition from mitochondria to hydrogenosomes (Stairs et al. 2021).

Environmentally important sulfate-reducing bacteria of the family Desulfobacteraceae contribute to carbon and sulfur cycling in marine sediments by coupling the oxidation of organic compounds to the reduction of sulfate to sulfide (Wöhlbrand et al. 2025). *A. flamelloides* houses Desulfobacteraceae-related symbionts within a specialized membrane organelle termed the symbiosome, which opens to the external seawater through deep membrane-lined pores. A reconstructed cell contained 183 intact symbionts (plus 2 degenerating), accounting for approximately 8% of total cell volume (Jerlström-Hultqvist et al. 2024). Remarkably, more recent reconstructions reveal that the symbiosome is not a collection of separate vesicles but a single, fully interconnected membrane compartment that spans up to 15% of the total cell volume, with a membrane surface area matching that of the plasma membrane (Jerlström-Hultqvist 2026). This architecture may provide the intracellular symbionts with access to environmental sulfate while they remain in close proximity to host hydrogenosomes. Hydrogenosomes generate cellular energy and release metabolic products, including molecular hydrogen (H<sub>2</sub>) and acetate, which can be utilized by the bacterial symbionts. This close association is consistent with a syntrophic partnership in which host-derived metabolites support bacterial sulfate-reducing metabolism (Stairs et al. 2021; Jerlström-Hultqvist 2026). Interestingly, the symbionts of *A. flamelloides* and *A. ignava* are both related to the Desulfobacteraceae but belong to different phylogenetic groups, suggesting that the symbionts were acquired independently rather than inherited from a single ancient symbiosis (Jerlström-Hultqvist et al. 2024).

## Bibliography

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